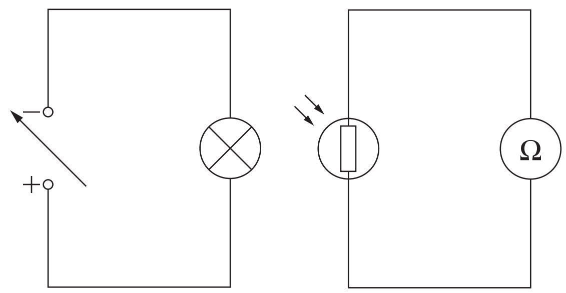
Examiner only

Answer **all** *questions.*

1. The following circuits are set up to investigate a light dependent resistor (LDR). The voltage of the power supply is changed to vary the power of the lamp to alter its brightness. The resistance of the LDR is measured with an ohmmeter (Ω) for each power of the lamp.



(a) (i) State **two** variables, **other than using the same components**, that should be controlled in this experiment. [2]

1. ....

2. ....

(ii) Explain how the design of the experiment could be improved to make the results more valid. [2]

....

....

....

(b) The results are shown in the table below.

Power of lamp (W)	Resistance of LDR (kΩ)
2	19.5
4	10.3
8	3.0
12	2.2
16	1.5
20	1.3
24	1.1

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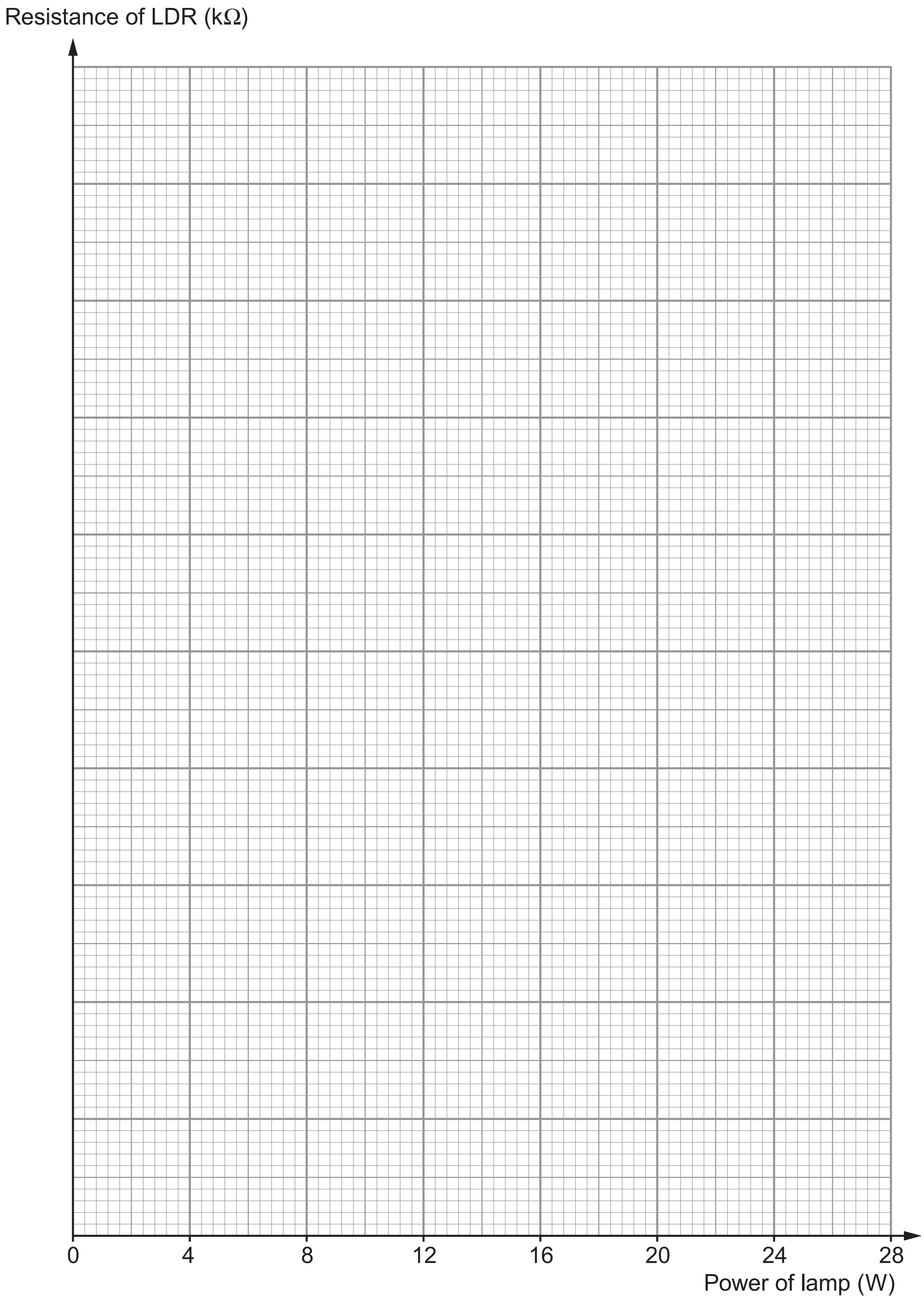
(i) Use the data to plot a graph on the grid below and draw a suitable line. [4]



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(ii) Use the graph to find the resistance of the LDR for a lamp power of 10 W. [1]

Resistance = ... Ω

(iii) It is suggested that when the lamp power doubles, the LDR resistance halves. Explain, using values from the table, to what extent this suggestion is true. [3]

....

....

....

(c) The LDR is connected into another circuit. The voltage across the LDR is 2.8 V and the current through it is 0.35 mA. Use an equation from page 2 to calculate the resistance of the LDR. [3]

Resistance = ... Ω

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